

VideoAnalysisJob		
UUID	job_id	PK
string	filename	
string	file_path	
string	pipeline_mode	
string	annotated_video_path	
string	status	
datetime	created_at	
datetime	started_at	
datetime	completed_at	
float	progress_percent	
int	total_frames	
int	processed_frames	
string	error_message	
json	component_failures	
json	metadata	

contains



Frame		
UUID	frame_id	PK
UUID	job_id	FK
int	frame_number	
int	timestamp_ms	
string	image_path	

produces

float	fps	
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has



Detection		
UUID	detection_id	PK
UUID	frame_id	FK
string	class_name	
float	confidence	
json	bbox_xyxy	
string	crop_image_path	



generates



FrameEmbedding		
UUID	embedding_id	PK
UUID	detection_id	FK
UUID	student_track_id	FK
string	model_name	
vector[768]	embedding_vector	
datetime	created_at	
int	embedding_age_seconds	



uses



StudentTrack		
UUID	track_id	PK
UUID	job_id	FK
int	tracking_id	
string	color_hex	
datetime	first_seen	
datetime	last_seen	
int	total_frames_visible	
json	track_metadata	



renders



BoundingBox		
UUID	bbox_id	PK
UUID	frame_id	FK
UUID	detection_id	FK
UUID	student_track_id	FK
json	xyxy	
string	color_hex	
string	tracking_id_text	

labels

